

Question:

A previously healthy 15-year-old girl is brought to the emergency department 24 hours after the onset of a severe headache. She returned from a 1-week camping trip 3 days ago; she went spelunking and swimming in a freshwater lake during the trip. She is agitated, uncooperative, and oriented only to person. Her temperature is 38.9°C (102°F), pulse is 112/min, respirations are 20/min, and blood pressure is 100/68 mm Hg. There are several crusted insect bites on her extremities. Neurologic examination shows diffuse hyperreflexia and an extensor plantar response bilaterally. Her neck is supple without lymphadenopathy. An MRI of the brain shows asymmetrical, bitemporal hyperintensities. A lumbar puncture is performed. Cerebrospinal fluid analysis shows: Opening pressure 150 mm H₂O Glucose 58 mg/dL Protein 108 mg/dL Leukocyte count 150/mm³ Segmented neutrophils 15% Lymphocytes 85% Erythrocyte count 25/mm³ Which of the following is the most likely causal pathogen?" Answer Choices: A. West Nile virus B. La Crosse virus C. Enterovirus D. Herpes simplex virus

Reasoning + Answer:

<think> Given the symptomatics including elevated cerebrospinal fluid protein and indicative MRI showing bitemporal hyperintensities, indicating presence of viral encephalitis, pointing towards potential Enterovirus infection. </think> <answer> Enterovirus </answer>

